

William Ayoade

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Education

University of California, San Diego

Mathematics-Computer Science
GPA - 3.825

Graduating 2026
San Diego, CA

Inderkum High School

GPA - 4.491(Weighted) / 3.873(Unweighted)
SAT - 1490 AP Scholar Award

Graduated 2023
Sacramento, CA

Projects

Custom Data Structures Library - Java

- Developed a comprehensive Java package of reusable data structures from scratch.
- Implemented core structures such as arraylists, linked lists, stacks, queues, hash maps, and binary search trees etc.
- Wrote extensive unit tests to ensure correctness and performance under edge cases.

Autocomplete Bot - Python

- Built an intelligent autocomplete system using n-gram models and likelihood estimation

Custom Malloc and Free Implementation - C

- Thread-safe memory management, with coalescing and best-fit algorithms to reduce fragmentation and improve performance.
- Achieved 30% faster allocations than glibc malloc() for specific workloads.

Customer Term Deposit Prediction Agent - Python, pgmpy, SendGrid API

- AI agent that predicts customer likelihood of subscribing to a term deposit using real-world bank marketing data of key features (age, job, education, etc)
- Utilizes Bayesian inference along with polynomial feature engineering and categorical variable encoding to enhance model performance
- Achieved 87.65% prediction accuracy

Finance Company Website - HTML/Javascript

- Website for a mock finance company, featuring multiple pages and interactive javascript

Skills

- Python
- Java
- C
- HTML/CSS
- Javascript
- C++
- React
- Excel

Experience

Volunteer & Planner

Joey's Food Locker

- Led food distribution stations
- Helped in providing food for over 4,000 people

Teachers Assistant

Inderkum High School

- Mentored and guided students
- Utilized leadership, management, and communication skills

Mathematics Tutor

Inderkum High School

- Gave students instruction and guidance on math topics